

Laboratory Exercise No. 2: Departmental LAN Segmentation and Dynamic Addressing

This exercise is a continuation of Lab Exercise 1.0, expanding the network to support multiple logical subnets and automated IP addressing for three distinct departments.

Objective:

Segment a 25-computer network into three departments: CICS, Cash, and Computer Service. Implement Inter-VLAN routing using a Multilayer Switch to allow different network addresses to communicate.

Configure Static IPv4 addresses for CICS and Cash departments.

Configure a DHCP Server to provide dynamic addressing for the Computer Service department.

Materials:

Cisco Packet Tracer Software.

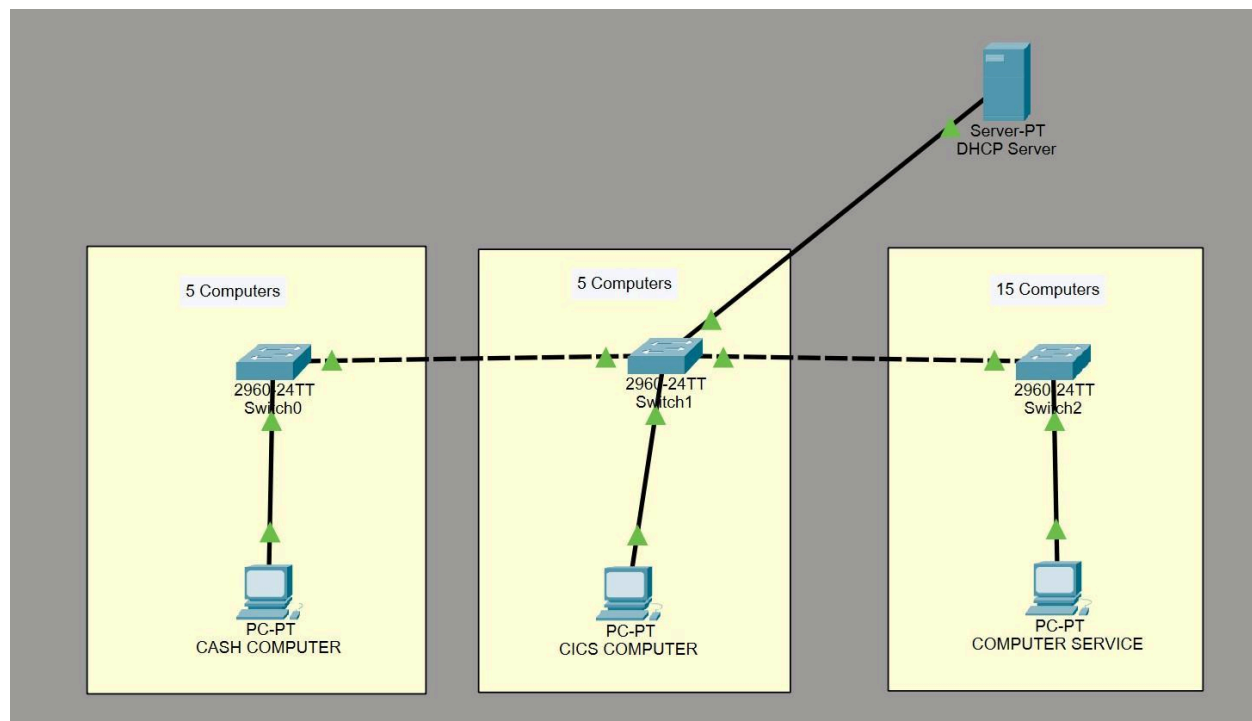
1 Multilayer Switch (Cisco 3560-24PS).

1 Standard Switch (Cisco 2960) to provide additional ports.

25 Generic PCs.

1 Generic Server.

Diagram:



Part 1: Topology Design and VLAN Assignment

1. Add Network Devices: Place three 2960 Switch as the access layer. Connect the switches together using a Copper Cross-over cable.
2. Add Host Computers:
 - CICS Department: 5 PCs (Network 192.168.1.0/24).
 - Cash Department: 5 PCs (Network 192.168.2.0/24).
 - Computer Service Department: 15 PCs (Network 192.168.3.0/24).
3. Physical Cabling: Use Copper Straight-Through cables to connect PCs and the Server to available FastEthernet ports on the switches.

Part 2: Manual Configuration (CICS & Cash Office)

For the 10 computers in these departments, you must configure the IP settings manually through the Desktop > IP Configuration tab.

Department
PC Range
IP Address Range
Subnet Mask
Default Gateway

CICS Dept

PC0 - PC4
IP: 192.168.1.10 - 192.168.1.14
SM: 255.255.255.0
GW: 192.168.1.1

Cash Dept.

PC5 - PC9
IP: 192.168.2.10 - 192.168.2.14
SM: 255.255.255.0
GW: 192.168.2.1

Part 3: DHCP Server Configuration (Computer Service Dept.)

The Computer Service Department (15 PCs) will receive addresses dynamically from a central server.

1. Static Server IP: Assign the Server a static IP of 192.168.3.254 and a gateway of 192.168.3.1.

2. Enable DHCP Service: Go to the Services tab and select DHCP.

3. Create Pool:

- Pool Name: Comp_Service_Pool.
- Default Gateway: 192.168.3.1.
- Start IP Address: 192.168.3.10.
- Subnet Mask: 255.255.255.0.
- Max Users: 50.
- Click Save and turn the Service On.

4. Client Configuration: On PC10 through PC24, go to IP Configuration and select the DHCP radio button. Verify they receive a 192.168.3.x address.

Part 5: Verification and Documentation

1. Ping Test: From a CICS PC (192.168.1.10), ping a Computer Service PC (192.168.3.10).

What is your observation?

2. Documentation Table: Create a table recording the Device Name, Department, IP Address, and Configuration Method (Static vs. DHCP).